

TPI Accessibility Standard

Version: 1.0
WCAG Level: AA (minimum), AAA (where possible)
Last Updated: 2025-01-XX

Overview

All TPI portals must meet WCAG 2.1 Level AA standards as a minimum. This document defines accessibility requirements and implementation guidelines.

WCAG Compliance

Level AA Requirements

Perceivable

- Text alternatives for images
- Captions for media
- Sufficient color contrast (4.5:1 for text)
- Resizable text (up to 200%)

Operable

- Keyboard accessible
- No seizure-inducing content
- Navigable with assistive technology
- Sufficient time limits

Understandable

- Readable text
- Predictable functionality
- Input assistance

Robust

- Compatible with assistive technology
 - Valid HTML
 - Proper ARIA usage
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Keyboard Navigation

Tab Order

Standard Navigation

- Tab: Move forward through focusable elements
- Shift+Tab: Move backward
- Order: Logical, top-to-bottom, left-to-right

Skip Links

- Position: First focusable element on page
- Target: Main content area
- Format: "Skip to main content"
- Visible: On focus only

Keyboard Shortcuts

Global Shortcuts

- Cmd/Ctrl+K: Open global search
- Escape: Close modal/drawer
- Enter: Activate button/link
- Space: Toggle checkbox/switch

Table Navigation

- Arrow keys: Navigate cells (if enabled)
- Enter: Activate row (open drawer)
- Space: Select row

Form Navigation

- Tab: Next field
- Shift+Tab: Previous field
- Enter: Submit form (if in form context)

Focus Management

Focus Indicators

- Visible: Always show focus outline
- Style: 2px solid primary color
- Contrast: Meets WCAG AA (3:1)
- Never remove: Focus styles are required

Focus Trapping

- Modals: Trap focus within modal
- Drawers: Trap focus within drawer
- Return focus: To trigger element on close

Focus Order

- Logical: Follows visual order
- Predictable: Same order every time
- Skip: Hidden elements not in tab order

Screen Reader Support

ARIA Labels

Required Attributes

- `aria-label`: For elements without visible text
- `aria-labelledby`: Link to visible label
- `aria-describedby`: Link to description
- `aria-live`: For dynamic content updates

Landmark Roles

- `role="main"`: Main content area
- `role="navigation"`: Navigation menus
- `role="banner"`: Header/top bar
- `role="contentinfo"`: Footer
- `role="complementary"`: Sidebar

Semantic HTML

Use Semantic Elements

- `<header>`, `<nav>`, `<main>`, `<footer>`
- `<section>`, `<article>`, `<aside>`
- `<button>`, `<a>`, `<input>`, `<select>`

Avoid

- `<div>` for buttons (use `<button>`)
- `<span>` for links (use `<a>`)
- Generic elements without roles

Announcements

Dynamic Updates

- `aria-live="polite"`: Non-urgent updates
- `aria-live="assertive"`: Urgent updates
- `aria-live="off"`: No announcements

Examples

- Toast notifications: `aria-live="polite"`
- Error messages: `aria-live="assertive"`
- Loading states: `aria-live="off"`

Color Contrast

Text Contrast

Normal Text (WCAG AA)

- Ratio: 4.5:1 minimum
- Use: Body text, labels, descriptions

Large Text (WCAG AA)

- Ratio: 3:1 minimum
- Size: 18pt+ or 14pt+ bold
- Use: Headings, large labels

Enhanced Text (WCAG AAA)

- Ratio: 7:1 minimum
- Use: When possible for better accessibility

Interactive Elements

Focus Indicators

- Ratio: 3:1 minimum against background
- Style: 2px solid outline
- Color: Primary color or high contrast

Buttons

- Text: 4.5:1 against button background
- Border: 3:1 against surrounding background

Status Colors

Not Color Alone

- Use: Color + icon + text
- Ensure: Information conveyed without color
- Test: Grayscale mode

Examples

- Status badge: Green + checkmark + "Active"
- Error message: Red + X icon + error text
- Warning: Yellow + warning icon + warning text

Form Accessibility

Labels

Required Labels

- Every input must have a label
- Format: `<label for="input-id">` or `aria-label`
- Visible: Labels should be visible

Help Text

- Link: `aria-describedby` to help text
- Position: Below field
- Format: Clear, concise instructions

Error Messages

Inline Errors

- Position: Below field
- Link: `aria-describedby` to error message
- Format: Clear, actionable error text
- Announce: To screen readers immediately

Error Summary

- Position: Top of form
- Link: To fields with errors
- Format: List of all errors
- Focus: Move focus to summary on submit

Required Fields

Indicators

- Visual: Asterisk (*) + "Required" text
- ARIA: `aria-required="true"`
- Format: "Field Name *"

Validation

- Real-time: Validate on blur
- Submit: Validate all fields on submit
- Clear: Clear errors when fixed

Images & Media

Alt Text

Decorative Images

- Format: `alt=""` (empty)
- Use: Icons, decorative graphics
- Avoid: Redundant text

Informative Images

- Format: `alt="[description]"`
- Content: Describe image content
- Context: Include relevant context

Functional Images

- Format: `alt="[function]"`
- Content: Describe function, not appearance
- Example: `alt="Close dialog"` not `alt="X icon"`

Captions & Transcripts

Video Content

- Captions: Required for all video
- Format: Synchronized captions
- Quality: Accurate, readable

Audio Content

- Transcripts: Required for audio-only
 - Format: Text transcript available
 - Link: Accessible from media player
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Responsive Design

Touch Targets

Minimum Size

- Size: 44x44px minimum
- Spacing: 8px between targets
- Use: Buttons, links, interactive elements

Mobile Considerations

- Larger: Increase size on mobile
- Spacing: More space between targets
- Avoid: Small, closely spaced targets

Viewport

Zoom Support

- Allow: Up to 200% zoom
- Layout: Remains usable at 200%
- Text: Remains readable

Orientation

- Support: Portrait and landscape
 - Layout: Adapts to orientation
 - No restriction: Don't lock orientation
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Testing

Automated Testing

Tools

- axe DevTools: Browser extension
- WAVE: Web accessibility evaluation
- Lighthouse: Accessibility audit

Frequency

- Pre-commit: Run automated checks
- CI/CD: Include in build pipeline
- Regular: Weekly automated scans

Manual Testing

Keyboard Testing

- Navigate: Entire interface with keyboard only
- Focus: All interactive elements focusable
- Order: Logical tab order

Screen Reader Testing

- Tools: NVDA (Windows), VoiceOver (Mac)
- Test: All major user flows
- Verify: Announcements are clear

Color Testing

- Grayscale: Test without color
- Contrast: Verify contrast ratios
- Colorblind: Test with colorblind simulators

User Testing

Involve Users

- Test: With actual assistive technology users
- Feedback: Regular accessibility reviews
- Iterate: Based on user feedback

Common Issues & Solutions

Issue: Missing Focus Indicators

Problem: Focus not visible **Solution:** Add visible focus outline (2px solid primary color)

Issue: Low Contrast Text

Problem: Text hard to read **Solution:** Increase contrast to 4.5:1 minimum

Issue: Missing Alt Text

Problem: Images not described **Solution:** Add descriptive alt text or empty alt for decorative

Issue: Keyboard Trapping

Problem: Focus escapes modal **Solution:** Implement focus trap in modal/drawer

Issue: Missing ARIA Labels

Problem: Screen readers can't identify elements **Solution:** Add appropriate ARIA attributes

Issue: Color-Only Indicators

Problem: Information only in color **Solution:** Add icon + text to convey information

Implementation Checklist

Page Level

- ☐ Semantic HTML structure
- ☐ Skip link present
- ☐ Landmark roles defined
- ☐ Page title descriptive
- ☐ Heading hierarchy logical

Navigation

- ☐ Keyboard accessible
- ☐ Focus indicators visible
- ☐ ARIA labels on navigation
- ☐ Current page indicated

Forms

- ☐ All inputs have labels
- ☐ Required fields indicated
- ☐ Error messages linked
- ☐ Help text available
- ☐ Validation accessible

Content

- ☐ Images have alt text
- ☐ Color contrast sufficient
- ☐ Text resizable
- ☐ Links descriptive
- ☐ Headings logical

Interactive Elements

- ☐ Buttons keyboard accessible
 - ☐ Focus management correct
 - ☐ ARIA attributes present
 - ☐ Status announced
 - ☐ Loading states announced
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Resources

Tools

- [axe DevTools](#)
- [WAVE](#)
- [Lighthouse](#)

Guidelines

- [WCAG 2.1 Guidelines](#)
- [ARIA Authoring Practices](#)
- [WebAIM](#)

Testing

- [NVDA Screen Reader](#)
 - [VoiceOver Guide](#)
 - [Color Contrast Checker](#)
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Related:

- [Content Style Guide](#)
- [Component Inventory](#)